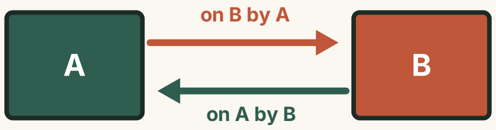
Newton’s Third Law

Every force belongs to an interaction between two objects.

ESSENTIAL QUESTION

If interaction forces are equal and opposite, how can either object accelerate?



LESSON MAP

Build the idea in four moves.

01

OBSERVE

A force never appears alone.

02

PAIR

Name both objects and both forces.

03

SEPARATE

Do not confuse pairs with balance.

04

APPLY

Use one free-body diagram at a time.

THE HABIT

For every force, ask: which object exerts it, and which receives it?

OBSERVE

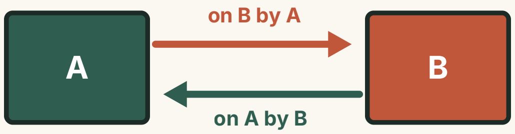
The rower moves because the water pushes back.

The oar pushes water backwards; the water pushes the oar and boat forwards.

1. **Boat acts on water**
   The oars accelerate water backwards.

2. **Water acts on boat**
   The water accelerates the boat forwards.

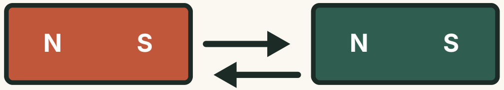
3. **One interaction**
   The two forces exist together.



OBSERVE

Contact is not required for an interaction.

Magnets, charged particles, and gravitating masses exert forces across space.



equal attraction, opposite directions

Forces reveal **interactions**, not necessarily touching.

EITHER WAY

Each partner pulls the other equally and oppositely.

MODEL

Every interaction produces a matched pair.

The force on A by B is equal in magnitude and opposite in direction to the force on B by A.

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

01 · Same magnitude

Both forces have identical size.

02 · Opposite direction

The minus sign records opposite vectors.

03 · Different objects

One acts on A; the other on B.

CHECK

A true third-law pair passes four tests.

Use these before calling two forces an action–reaction pair.

01 · Same interaction

Both forces are the same physical type — contact, gravity, electric.

02 · Different objects

The forces act on the two partners, never on one object.

03 · Equal & opposite

Equal magnitude, opposite direction.

04 · Simultaneous

They begin, change, and disappear together.

COMPARE

Third-law pairs are not balanced forces.

The difference is not size or direction — it is which object receives each force.

INTERACTION PAIR

Forces act on **different** objects — they cannot cancel on a single free-body diagram.

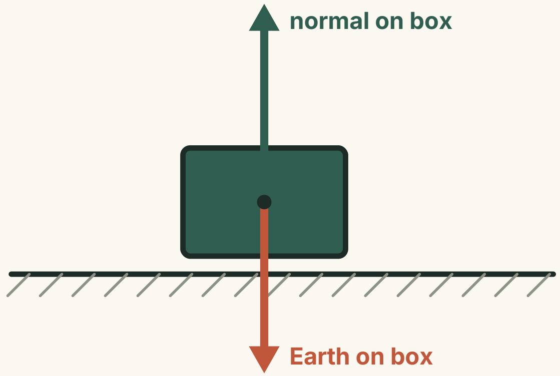
BALANCED FORCES

Forces act on the **same** object — they may add to zero, giving zero acceleration.

Draw one object at a time.

Only forces acting on the chosen object belong on its diagram.

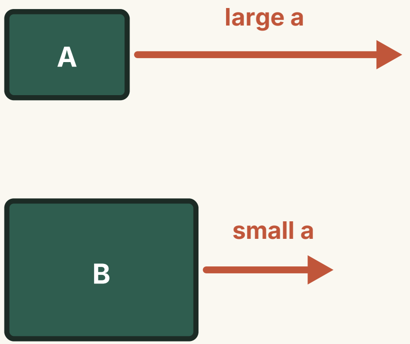
1. **Choose the object**
   Here, the box is the system.
2. **List forces on it**
   The table's normal force and Earth's gravity act on the box.
3. **Keep partners elsewhere**
   The box-on-table and box-on-Earth forces belong to other diagrams.



DERIVE

Equal forces can produce unequal accelerations.

The third law fixes the force pair; the second law connects each force to its object's mass.



same force, unequal acceleration

SECOND LAW ON EACH BODY

$$a = \frac{F}{m}$$

The lighter object accelerates more, even though both feel the same force magnitude.

APPLY

Motion follows from pushing on something else.

Walking, rowing, and rockets use the same interaction logic.

01 · Walking

You push the ground back; static friction pushes you forwards.

02 · Rowing

The oars push water back; water pushes the boat forwards.

03 · Rocket

The engine pushes exhaust back; exhaust pushes the rocket forwards.

WORKED EXAMPLE

Buoyancy transfers a force to the scale.

A suspended 10 N block is partly immersed. The spring scale now reads 6 N.

1. **Find the buoyant force**

   $F_b = 10 - 6 = 4\ \mathrm{N}$

2. **Use the third law**

   $\Delta(\text{scale}) = +4\ \mathrm{N}$

THE PAIRING

The liquid pushes the block up (buoyancy); by the third law the block pushes the liquid down, and that force reaches the scale.

COMMON TRAPS

Three tempting statements are wrong.

Repair each by naming the objects carefully.

01 · “The bigger object exerts more.”

Wrong. Interaction forces are equal; the smaller mass usually accelerates more.

02 · “The reaction happens after.”

Wrong. Both forces exist and change at the same time.

03 · “The pair cancels.”

Wrong for one object — the two forces act on different objects.

Use a reliable four-step method.

This route prevents nearly every action–reaction mistake.

01 · Name the interaction

Contact? Gravity? Electric? Magnetic?

02 · Name both forces

“Force on A by B” and “force on B by A”.

03 · Choose one object

Draw only the forces acting on it.

04 · Apply the second law

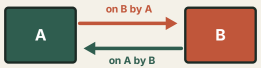
Use the net force on the chosen object.

SUMMARY

A force is one half of an interaction — its partner is equal, opposite, and on the other object.

THE INTERACTION LAW

$$\vec{F}_{AB} = -\vec{F}_{BA}$$



Separate the objects before analysing motion.

Equal and opposite forces don’t cancel — they act on different objects, so each can accelerate.